Multiplying & Dividing Algebraic Fractions

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Multiplying & dividing algebraic fractions

How do I multiply algebraic fractions?

- **STEP 1**
  Simplify both fractions first by **fully factorising**
  - E.g. $\frac{x}{3x+6} \times \frac{2x+4}{x+7} = \frac{x}{3(x+2)} \times \frac{2(x+2)}{x+7}$
- **STEP 2**
  Cancel any common factors on top and bottom (from either fraction)
  - E.g. $\frac{x}{3\cancel{(x+2)}} \times \frac{2\cancel{(x+2)}}{x+7} = \frac{x}{3} \times \frac{2}{x+7}$
- **STEP 3**
  Multiply the **tops** together
  Multiply the **bottoms** together
  - E.g. $\frac{2x}{3(x+7)}$
- **STEP 4**
  Check for any further factorising and cancelling
  - E.g. $\frac{2x}{3(x+7)}$ has no common factors so is in its simplest form

Help me understand

Worked Example

Write the following as a single fraction in its simplest form

$$\frac{2x^2-11x+12}{x^2+3x-28} \times \frac{x^2-5x-14}{9-4x^2}$$

Answer:

Factorise the four quadratics

$2x^2-11x+12$

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$2x^2 - 3x - 8x + 12$$
$$x(2x-3) - 4(2x-3)$$
$$(2x-3)(x-4)$$

$x^2 + 3x - 28$

- Two numbers that multiply to give -28 and add to give 3
- 7 and -4

$$(x+7)(x-4)$$

$x^2 - 5x - 14$

- Two numbers that multiply to give -14 and add to give -5
- 2 and -7

$$(x+2)(x-7)$$

$9 - 4x^2$

- Difference of two squares
- $3^2 - (2x)^2$

$$(3-2x)(3+2x)$$

Write the expression using the factors

$$\frac{(2x-3)(x-4)}{(x+7)(x-4)} \times \frac{(x+2)(x-7)}{(3-2x)(3+2x)}$$

Rewrite $2x - 3$ as $-(3-2x)$ and cancel common factors

$$\frac{-\cancel{(3-2x)}\cancel{(x-4)}}{(x+7)\cancel{(x-4)}} \times \frac{(x+2)(x-7)}{\cancel{(3-2x)}(3+2x)}$$
$$\frac{-1}{(x+7)} \times \frac{(x+2)(x-7)}{(3+2x)}$$

Write as a single fraction

- $(x+7)$ and $(x-7)$ are not common factors
- You can multiply $(x-7)$ by -1 to get rid of the negative sign in front of the front
- Leave the expressions factorised

$$\frac{-(x+2)(x-7)}{(x+7)(3+2x)} \text{ or } \frac{(x+2)(7-x)}{(x+7)(3+2x)}$$

How do I divide algebraic fractions?

- **Flip** (find the reciprocal of) the **second** fraction and replace ÷ with ×
  - So $\div \frac{a}{b}$ becomes $\times \frac{b}{a}$
  - E.g. $\frac{3x-12}{x} \div \frac{2x+8}{x+3} = \frac{3x-12}{x} \times \frac{x+3}{2x+8}$
- Then follow the same rules for **multiplying** two fractions

$x-4 \qquad x^2-16$

fraction.

Answer:

Division is the same as multiplying by the reciprocal (the fraction flipped)

$$\frac{x+3}{x-4} \div \frac{2x+6}{x^2-16} = \frac{x+3}{x-4} \times \frac{x^2-16}{2x+6}$$

Factorise all numerators and denominators to see which factors cancel out
You need to use the difference of two squares,
$x^2-4^2=(x-4)(x+4)$

$$\frac{x+3}{x-4} \times \frac{x^2-16}{2x+6} = \frac{\cancel{x+3}}{\cancel{x-4}} \times \frac{\cancel{(x-4)}(x+4)}{2\cancel{(x+3)}}$$

Multiply the remaining numerators and denominators together

$$\frac{1\times(x+4)}{1\times 2} = \frac{x+4}{2}$$

Check to see if you missed any terms that are the same on the top and bottom that could be cancelled

$\frac{x+4}{2}$ is already in its simplest form

$$\frac{x+4}{2}$$

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